COPELAND

Software Engineer Take-Home Assignment

At Copeland, we'd like to learn more about your technical skills and how you approach real-world problems. Rather than thinking of this as a pass/fail exercise, treat it as an opportunity to showcase your abilities and how you build software.

We've designed this assignment to be practical, open-ended, and (hopefully) enjoyable. You're free to use any language, frameworks, tools, or third-party libraries you prefer.

We estimate this should take approximately **3–8 hours** to complete. You're welcome to work on it at your convenience.

Please submit your completed assignment by the agreed-upon date, or let us know if you need additional time. If anything is unclear, don't hesitate to reach out with questions.

What We're Looking For

We value code that is:

- Clear and easy to understand
- Well-structured and maintainable
- Thoughtfully designed

While not required, we also appreciate when candidates incorporate practices they would normally use in production environments, such as:

- Tests
- Build scripts (e.g., Makefile or equivalent)
- Tooling and automation
- Documentation
- Source control with meaningful commits

Assignment

Build an application that allows users to retrieve **current weather conditions** by:

- City name
- ZIP code
- Geographic coordinates (GPS / reverse geolocation)

Use the OpenWeatherMap API (https://openweathermap.org/api)

Your application should be designed in a way that makes it easy to extend with new features and straightforward to test and maintain.

Deployment Requirement

In addition to the application itself, please:

- Provision and configure a **private network environment** using tools of your choice
- Deploy and run your application within this environment
- Restrict access so the application is only reachable via a **VPN connection**

Include:

- A VPN configuration file for accessing the network
- Clear instructions for connecting to the VPN and accessing the application
- Confirmation that the application is **not accessible outside the VPN** (e.g., fails with NXDOMAIN or similar)

Submission Requirements

Please include:

- Source code
- Setup and run instructions
- Any relevant documentation
- Examples of how to use the application

API Access

You can generate your own OpenWeatherMap API key (free), though activation may take a few days.

In the meantime, you may use the following token:

cf002751564a4c78f5f7ed479f1b9ba3

Use of AI Tools

We recognize that modern development often includes the use of AI-assisted tools (e.g., ChatGPT, GitHub Copilot, etc.), and you are welcome to use them as part of this assignment.

That said, our goal is to understand **your** problem-solving approach and technical judgment. If you choose to use AI tools, please follow these guidelines:

- Be prepared to **explain and defend** any code you submit

- Ensure you **fully understand** all parts of your solution
- Avoid blindly copying or relying on generated code without review
- Use AI as a **tool to assist**, not replace, your own thinking

If AI tools are used, include a brief note describing:

- **Where** you used them (e.g., debugging, scaffolding, documentation)
- **How** they influenced your solution
- Any **key prompts, instructions, or techniques** you used to guide the output

Submissions that appear to be largely AI-generated without clear understanding or ownership may be viewed less favorably.